THRIFTKART / SEMESTER III

Test Plan & Evidence

BVA, equivalence classes, decision tables and metrics

Focus	Listing moderation + price/condition rules
Coverage	12 functional trace links
Critical techniques	BVA • Equivalence • Decision tables • Integration
Metrics	Defect density + DRE
Evidence status	Designed; execution values to be filled from actual run

1. Test Objectives

- Verify listing validation catches invalid price and condition values.
- Verify no unapproved listing is publicly visible.
- Verify a listing cannot receive conflicting active reservations.
- Verify the list-to-sale-to-handover workflow preserves valid data relationships.
- Produce a defect log from actual execution so DRE and defect density can be calculated without fabricated results.

2. Boundary Value Analysis — Price

Case	Input	Expected	Reason
TC-P01	0	Reject	Just below lower boundary
TC-P02	1	Accept	Lower boundary
TC-P03	2	Accept	Just above lower boundary
TC-P04	19,999	Accept	Just below upper boundary
TC-P05	20,000	Accept	Upper boundary
TC-P06	20,001	Reject	Just above upper boundary

3. Boundary Value Analysis — Condition

Case	Input	Expected	Reason
TC-C01	0	Reject	Just below lower boundary
TC-C02	1	Accept	Lower boundary
TC-C03	2	Accept	Just above lower boundary
TC-C04	4	Accept	Just below upper boundary
TC-C05	5	Accept	Upper boundary
TC-C06	6	Reject	Just above upper boundary

4. Equivalence Classes

Field	Class	Values	Expected
Price	Invalid low	≤ 0	Reject
Price	Valid	1–20,000	Accept
Price	Invalid high	≥ 20,001	Reject
Condition	Invalid low	≤ 0	Reject
Condition	Valid	1–5	Accept
Condition	Invalid high	≥ 6	Reject

5. Decision Table — Listing Moderation

Rule	Required fields	Price valid	Condition valid	Content acceptable	Expected
R1	Y	Y	Y	Y	Approve
R2	N	Y	Y	Y	Reject
R3	Y	N	Y	Y	Reject
R4	Y	Y	N	Y	Reject
R5	Y	Y	Y	N	Reject

6. Decision Table — Reservation / Price Rule

Rule	Listing available	Active reservation exists	Price valid	Expected
R1	Y	N	Y	Create reservation
R2	N	N	Y	Reject request
R3	Y	Y	Y	Reject request
R4	Y	N	N	Reject listing/request

7. Functional Trace Test Cases

Test ID	FR	Test purpose
TC-01	FR-01	Register/login with valid and invalid university account credentials.
TC-02	FR-02	View and update profile data while authenticated.
TC-03	FR-03	Create a draft listing with all required fields.
TC-04	FR-04	Run price/condition BVA and verify reject/accept boundaries.
TC-05	FR-05	Submit listing, approve/reject it, verify public visibility rule.
TC-06	FR-06	Search by title and filter by condition/price.
TC-07	FR-07	Open listing detail and verify seller, price, condition and pickup.
TC-08	FR-08	Open listing-scoped chat and send/receive messages.
TC-09	FR-09	Create reservation and attempt a conflicting second reservation.
TC-10	FR-10	Record payment method/status and verify transaction audit record.
TC-11	FR-11	Record handover and complete transaction; verify listing Sold.
TC-12	FR-12	Submit report and resolve it through admin queue.

8. Defect Log Template

Defect ID	Test ID	Severity	Status	Root cause	Fix / evidence
D-001					
D-002					
D-003					

Defect ID	Test ID	Severity	Status	Root cause	Fix / evidence
D-004					

9. Metrics

Defect Density

Defect Density = Confirmed defects / KLOC. Populate only after the test log contains executed results and the project has a measured code size.

Defect Removal Efficiency (DRE)

DRE = Defects removed before release / (Defects removed before release + defects found after release) × 100. Do not invent a numerical result; record the executed evidence first.

10. Exit Criteria

- All Must FRs executed at least once.
- All BVA boundary cases executed.
- Moderation decision table executed against each rule.
- No open critical defect remains at release.
- Defect density and DRE are calculated from the final defect/test log.